CMPE2150 SA 05

Relay and SSR Circuits

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Omron G2R-2-DC12

The first set of questions in this assessment are based upon your knowledge of transistor-controlled relay circuits and information provided in the manufacturer's datasheet for the relay in your kit, the Omron G2R-2-DC12.

Note that this datasheet covers a number of devices, so you need to make sure you're accessing the correct part number when accessing the specifications.

1. Dielectric strength refers to _______________

    a. the strength of a dielectric material, expressed as a function of aging due to repetitive arcing between terminal contacts.
    b. the maximum voltage that may be applied across the terminals of a relay while they are open.
    c. the maximum voltage that may be applied between the coil and the contacts before a resulting electric current will flow between them.
    d. a measure of the capacitance of the polarizing electrostatic field that develops between switch contacts while they are open.

2. What is the basic dielectric rating of the G2R-2-DC12 relay, in volts? _______________

3. The G2R-2-DC12 relay in your kit has a fully-sealed enclosure. ________________ (True or False)
4. What type of protection does the enclosure of the G2R-2-DC12 relay provide?
   a. prevents ingress of solder flux during soldering
   b. prevents damage due to corrosive atmospheric gases
   c. prevents protection from electromagnetic field flux
   d. prevents influx of cleaning solvent during immersion cleaning
5. How many circuits can the G2R-2-DC12 control simultaneously? ________________
6. For the G2R-2-DC12 relay, how many circuit elements can each relay switch be connected to? ________________
7. Which statement describes the operation of the contacts in the G2R-2-DC12 relay?
   a. all contacts are normally open
   b. all contacts are normally closed
   c. each switch has a normally-open contact and a normally-closed contact
8. What is the nominal coil voltage for the G2R-2-DC12 relay? ________________
9. For the G2R-2-DC12 relay, what type of power should be supplied to the relay coil? ________________ (AC or DC)
10. For the G2R-2-DC12 relay, once the relay is activated, what can the voltage be reduced to without the relay releasing, in volts? ________________
11. What is the coil current rating of the G2R-2-DC12 relay, in milliamps? ________________
12. What is the rated coil resistance of the G2R-2-DC12 relay? ________________

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Answers

Omron G2R-2-DC12

1. the maximum voltage that may be applied between the coil and the contacts before a resulting electric current will flow between them.
2. 5000V
3. False
4. prevents ingress of solder flux during soldering
5. 2
6. 2
7. each switch has a normally-open contact and a normally-closed contact
8. 12V
9. DC
10. 1.8V
11. 43.6mA
12. 275Ω

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